INDEPENDENT PROJECT DESIGN PORTFOLIO

Inventory and Sales Management System

Relational Database, ERD and AdminLTE 3 Interface Design

Student Name: _____

Registration Number: _____

Course / Class: _____

Instructor: _____

Submission Date: _____

Note: This is a design portfolio. Replace the illustrative values with your own if required, and make sure you can explain the design during the supervised review.

1. Project Selection and Brief

Selected option: B — Inventory and Sales.

Problem statement

Small shops need a reliable way to organize product details, categories, suppliers, stock quantities and sales. Keeping these records manually can cause stock-count errors, difficulty identifying low-stock products, and slow sales-history reviews.

Purpose

The proposed system will centralize product and sales information and provide staff with a clear dashboard and searchable lists.

Intended users

Administrator: manages system records and user access. Shop staff: searches products, records sales and checks stock. A simplified version may use one authorized staff role for the assignment.

Core features

Manage categories, suppliers and products; display current stock and low-stock products; record sales and their line items; calculate sale totals; review recent sales and basic summaries.

Boundaries and assumptions

This portfolio covers design only, not a deployed application. It does not include barcode scanning, online payments, tax integration, multiple branches or automatic supplier ordering. Prices are stored in Tanzanian shillings (TZS). Each product has one main supplier in this simplified model. A sale contains one or more sale items.

2. Project Scope

Included in scope	Outside the current scope
Product, category and supplier records; stock-level visibility; recording sales and sale items; recent activity; basic sales and low-stock summaries.	Barcode scanning; online payment gateway; tax integration; multiple branches; automated purchasing; live deployment and database implementation.

3. Relational Database Design and Data Dictionary

The proposed database uses five tables. Primary keys identify records. Foreign keys connect related tables and preserve referential integrity. Data types are illustrative and suitable for a relational database such as MySQL.

Categories

Purpose: Groups products into types.

Fields and data types: category_id INT PK; category_name VARCHAR(80) NOT NULL UNIQUE; description VARCHAR(255) NULL

Rules / constraints: Category name must be unique.

Suppliers

Purpose: Stores supplier contact details.

Fields and data types: supplier_id INT PK; supplier_name VARCHAR(120) NOT NULL; phone VARCHAR(25) NULL; email VARCHAR(120) NULL; address VARCHAR(255) NULL

Rules / constraints: Email is optional; validate format when provided.

Products

Purpose: Stores sellable product and stock details.

Fields and data types: product_id INT PK; sku VARCHAR(40) NOT NULL UNIQUE; product_name VARCHAR(120) NOT NULL; category_id INT FK; supplier_id INT FK NULL; unit_price DECIMAL(12,2) NOT NULL; quantity_in_stock INT NOT NULL DEFAULT 0; reorder_level INT NOT NULL DEFAULT 0; description VARCHAR(255) NULL; is_active BOOLEAN NOT NULL DEFAULT TRUE

Rules / constraints: unit_price >= 0; quantity_in_stock >= 0; reorder_level >= 0. category_id references Categories. supplier_id references Suppliers and may be NULL.

Sales

Purpose: One transaction header per sale.

Fields and data types: sale_id INT PK; sale_datetime DATETIME NOT NULL; staff_name VARCHAR(120) NOT NULL; payment_method VARCHAR(30) NOT NULL; sale_status VARCHAR(20) NOT NULL DEFAULT 'COMPLETED'; total_amount DECIMAL(12,2) NOT NULL DEFAULT 0

Rules / constraints: total_amount >= 0. payment_method should be one of CASH, MOBILE_MONEY, CARD, OTHER. Status may be COMPLETED or VOID.

SaleItems

Purpose: Products and quantities included in each sale.

Fields and data types: sale_item_id INT PK; sale_id INT FK; product_id INT FK; quantity INT NOT NULL; unit_price DECIMAL(12,2) NOT NULL; line_total DECIMAL(12,2) NOT NULL

Rules / constraints: quantity > 0; unit_price >= 0; line_total = quantity × unit_price. sale_id references Sales; product_id references Products. Save the selling price at sale time.

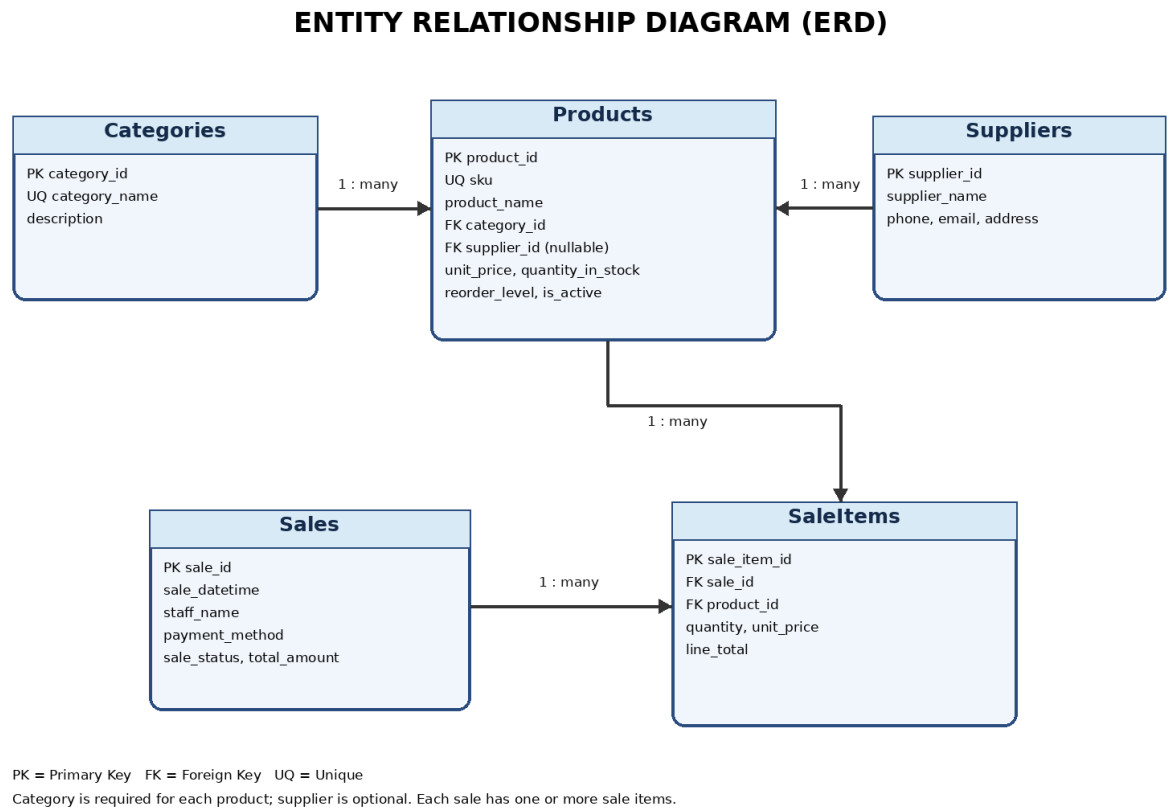
4. Relationships, Cardinality and Optionality

Relationship	Cardinality	Explanation
Categories to Products	One-to-many (1:M)	Each product must belong to one category. A category can contain zero or many products.
Suppliers to Products	One-to-many (1:M)	A product may have zero or one main supplier; a supplier can be linked to zero or many products.
Sales to SaleItems	One-to-many (1:M)	Each sale must contain one or more sale items. Each sale item belongs to exactly one sale.
Products to SaleItems	One-to-many (1:M)	A product can appear in zero or many sale items. Each sale item refers to exactly one product.

Optionality summary: every Product must reference one Category; a Product may have no Supplier or one Supplier. A Category or Supplier may exist before any products are linked to it. Each Sale must have one or more SaleItems. A Product may not yet have appeared in a sale.

5. Entity Relationship Diagram (ERD)

The diagram below shows the five entities, their primary and foreign keys, and the one-to-many relationships.



Legend: PK = Primary Key; FK = Foreign Key; UQ = Unique. The supplier reference in Products is nullable.

6. Relational Schema Outline

• Categories(category_id PK, category_name UNIQUE NOT NULL, description)

• Suppliers(supplier_id PK, supplier_name NOT NULL, phone, email, address)

• Products(product_id PK, sku UNIQUE NOT NULL, product_name NOT NULL, category_id FK → Categories.category_id, supplier_id FK → Suppliers.supplier_id NULL, unit_price, quantity_in_stock DEFAULT 0, reorder_level DEFAULT 0, description, is_active DEFAULT TRUE)

• Sales(sale_id PK, sale_datetime, staff_name, payment_method, sale_status DEFAULT 'COMPLETED', total_amount DEFAULT 0)

• SaleItems(sale_item_id PK, sale_id FK → Sales.sale_id, product_id FK → Products.product_id, quantity, unit_price, line_total)

7. Illustrative Sample Records

The records below are examples only; they are not real business records.

product_id	sku	product_name	category_id	supplier_id	unit_price	quantity_in_stock	reorder_level
1	P001	USB Mouse	1	1	15000.00	4	10
2	P002	Keyboard	1	1	25000.00	22	5
3	P003	HDMI Cable	2	2	10000.00	3	8

sale_id	sale_datetime	staff_name	payment_method	sale_status	total_amount
104	2026-10-18 10:30	Sample Staff	CASH	COMPLETED	85000.00

For SaleItems, each line records sale_id, product_id, quantity, unit_price and line_total. The sale total should equal the sum of its line totals.

8. AdminLTE 3 Interface Plan

The interface is designed with a consistent header, sidebar navigation, page titles, breadcrumbs, cards, tables, form controls and feedback. The wireframes below are labelled planning sketches rather than screenshots of a running application.

8.1 Main Navigation

Navigation item	Purpose
Dashboard	Stock summaries, today's sales and recent activity.
Products	Search, view, add and edit product records.
Categories	Maintain product categories.
Suppliers	Maintain supplier contact details.
Sales	Review sales and their item details.
Reports	Review basic stock and sales summaries.

8.2 Screen A — Dashboard Wireframe

Wireframe area	Planned content
HEADER: Inventory & Sales Management System	User menu | Notifications
SIDEBAR	Dashboard | Products | Categories | Suppliers | Sales | Reports
BREADCRUMB	Dashboard > Home
SMALL BOX 1	TOTAL PRODUCTS: 128
SMALL BOX 2	STOCK UNITS: 845
SMALL BOX 3	TODAY'S SALES: TZS 450,000
SMALL BOX 4	LOW-STOCK ITEMS: 7
CARD: RECENT SALES	Sale ID | Date | Total | Status
Example rows	S-104 | 18 Oct | 85,000 | Completed; S-103 | 18 Oct | 42,000 | Completed
CARD: LOW-STOCK ITEMS	Product | Quantity | Reorder level
Example rows	USB Mouse | 4 | 10; Keyboard | 2 | 5; HDMI Cable | 3 | 8
FOOTER	System name / version

Annotations: 1) Header and user menu. 2) Sidebar provides module navigation. 3) Breadcrumb shows location. 4) Four small boxes summarize product and sales data. 5) Recent Sales card displays latest transactions. 6) Low Stock Items card identifies products needing attention. All displayed figures are illustrative.

8.3 Screen B — Product List Wireframe

Wireframe area	Planned content
PAGE TITLE	Products
BREADCRUMB	Home > Products
SEARCH INPUT	Search by product name or SKU...
FILTERS	Category [Select v] | Stock status [Select v] | [Search]
PRIMARY BUTTON	[+ Add Product]
RESPONSIVE TABLE	SKU | Product | Category | Supplier | Qty | Unit Price | Status | Actions
EXAMPLE ROW 1	P001 | USB Mouse | Accessories | ABC Supplies | 4 | 15,000 | LOW STOCK | View / Edit
EXAMPLE ROW 2	P002 | Keyboard | Accessories | ABC Supplies | 22 | 25,000 | IN STOCK | View / Edit

Wireframe area	Planned content
EXAMPLE ROW 3	P003 | HDMI Cable | Cables | Tech Distributors | 3 | 10,000 | LOW STOCK | View / Edit
TABLE FOOTER	Showing 1–10 of 128 products | [Previous] 1 2 3 [Next]

Annotations: 1) Search by name or SKU. 2) Filters narrow results by category and stock condition. 3) Add Product opens the entry form. 4) Responsive table displays records. 5) Badges show stock status. 6) View/Edit actions open the selected record. 7) Pagination handles long lists.

8.4 Screen C — Add/Edit Product Wireframe

Field / area	Label or control
PAGE TITLE	Add / Edit Product
BREADCRUMB	Home > Products > Add Product
Product Name *	[Enter product name]
SKU / Product Code *	[Enter unique SKU]
Category *	[Select category v]
Main Supplier	[Select supplier v]
Unit Price (TZS) *	[0.00]
Quantity in Stock *	[0]
Reorder Level *	[0]
Description	[Optional notes]
ACTIONS	[Save Product] [Cancel]
FEEDBACK	Alert: Product saved successfully / Please correct highlighted fields.

Annotations: fields marked * are required. SKU must be unique. Category is selected from Categories. Supplier may be left blank. Price, stock quantity and reorder level must not be negative. Save should validate inputs and show an alert; Cancel returns to the list without saving.

9. AdminLTE Component-use Matrix

No.	Component	Screen	Purpose	Related data / task
Small Boxes / Info Boxes	Dashboard	Show key counts: products, stock units, today's sales and low-stock items.	Products, stock and sales summaries	
Sidebar Navigation Menu	All screens	Provide consistent links to Dashboard, Products, Categories, Suppliers, Sales and Reports.	Navigation between modules	
Breadcrumbs	Product list and product form	Show the current page and its parent location.	Current screen / navigation	
Responsive Table	Product list	Present product records in clear columns and adapt to narrower screens.	SKU, name, category, supplier, quantity, price and status	
Search Form / Input Group	Product list	Find a product by name or SKU.	Product search	
Select /Filter Controls	Product list	Filter records by category or stock condition.	Category and stock status	
Badges	Product list and dashboard	Display IN STOCK or LOW STOCK status at a glance.	Quantity compared with reorder level	
Buttons / Button Groups	List and product form	Provide Add Product, View, Edit, Save and Cancel actions.	Create, view and update actions	
Form Controls	Add/Edit Product	Collect product name, SKU, category, supplier, price and stock levels.	Product record entry	
Alerts	Add/Edit Product	Show a success message or explain validation errors.	Save feedback	
Cards	Dashboard	Group recent sales and low-stock items under clear headings.	Recent transactions and warnings	
Pagination	Product list	Split a large product list into pages.	Long result sets	

No.	Component	Screen	Purpose	Related data / task
Modal / Confirmation Dialog	Product list	Ask for confirmation before deactivating a product.	Safe deactivation	

10. Explanation of Important Design Choices

• Separate Categories and Suppliers tables reduce repeated text and help keep names consistent.

• Products stores category_id and supplier_id as foreign keys rather than repeating full category and supplier details in every product row.

• Sales and SaleItems are separate because one sale can contain multiple products. SaleItems stores the unit price at the time of sale so a later price change does not rewrite sales history.

• Quantity, unit price and reorder level have validation rules to prevent negative values. SKU and category_name are unique to avoid duplicate identifiers or category names.

• The Dashboard summarizes data, while the Product List supports search and actions. The Add/Edit form uses the same field names and rules as the Products table, keeping the interface aligned with the data model.

• Products that appear in sales history should normally be deactivated rather than permanently deleted, so historical sale details remain understandable.

11. Final Design Review Checklist

■ One project option and its scope are stated.

■ Data dictionary includes table purposes, fields, data types, keys and rules.

■ ERD shows entities, keys, labelled relationships, cardinality and optionality.

■ Relational schema matches the ERD and data dictionary.

■ Wireframes include Dashboard, List and Entry/Edit screens.

■ At least eight distinct AdminLTE 3 components are mapped to screens and tasks.

■ Screen labels and fields match the proposed data design.

■ Examples are labelled illustrative, and this submission is clearly described as design-only.

12. Conclusion

This portfolio proposes a coherent Inventory and Sales Management System design. The relational schema organizes products, categories, suppliers, sales and sale items. The ERD explains how the records connect, while the AdminLTE 3 wireframes show how users would navigate, search, review stock and enter product data. The proposed interface and database use matching fields and validation rules.